Published: 08 November 2024

IoTML 2024: 2024 4th International Conference on Internet of Things and Machine Learning
August 9 - 11, 2024
Nanchang, China

RESEARCH-ARTICLE

A text restoration model for ancient texts based on pre-trained language model RoBERTa

ZHONGYU GU, North China University of Technology, Beijing, China

YANZHI GUAN, North China University of Technology, Beijing, China

SHUAI ZHANG, North China University of Technology, Beijing, China

IoTML '24: Proceedings of the 2024 4th International Conference on Internet of Things and Machine Learning (August 2024)
https://doi.org/10.1145/3697467.3697610
ISBN: 9798400710353

A text restoration model for ancient texts based on pre-trained language model RoBERTa

Zhongyu Gu*
North China University of Technology, School of Mechanical and Materials Engineering
China
guzhongyu0614@163.com

Yanzhi Guan
North China University of Technology, School of Mechanical and Materials Engineering
China
gyz@ncut.edu.cn

Shuai Zhang
North China University of Technology, School of Mechanical and Materials Engineering
China
769885968@qq.com

Abstract

We propose an Ancient Chinese Text Restoration Model (ACTRM) that leverages the pre-trained RoBERTa model. This model is designed to continuously assimilate semantic information from the surrounding context to reconstruct missing text segments. The model's predictive capabilities are notably enhanced through the incorporation of phrase-level semantics within the input embedding layer and the application of a KAN network in the prediction layer. The primary objective of this model is to augment the workflow of specialists engaged in the restoration of ancient texts. To facilitate the training of the model, a dataset comprising ancient Chinese texts across diverse subjects—including science, technology, geography, history, and science fiction—has been developed. The model achieves an accuracy rate of 90.6% for Top-5 predictions and 76.7% for individual word predictions within this dataset, while also ensuring semantic coherence and fluency in the reconstructed text. These findings substantiate the model's efficacy as a supportive tool for the restoration of ancient texts, potentially enhancing both the efficiency and accuracy of such restoration efforts.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Natural language processing.

Keywords

Text infilling, pretrained model, phrase-level semantics, KAN network

ACM Reference Format:
Zhongyu Gu, Yanzhi Guan, and Shuai Zhang. 2024. A text restoration model for ancient texts based on pre-trained language model RoBERTa. In *2024 4th International Conference on Internet of Things and Machine Learning (IoTML 2024), August 09–11, 2024, Nanchang, China.* ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/3697467.3697610

*Corresponding author.

IoTML 2024, August 09–11, 2024, Nanchang, China

ACM ISBN 979-8-4007-1035-3/24/08
https://doi.org/10.1145/3697467.3697610

1 Introduction

In addition to its significant academic value, ancient literature is a valuable source of cultural relics and a witness to historical development and cultural inheritance. The original text of ancient books must be preserved under special circumstances because they were written a long time ago and are difficult to read or research. However, because digital technology is developing quickly these days, the ancient books will be converted into digital texts that will allow their research and study to be fully appreciated. However, over time, the paper has aged, naturally eroded, and undergone poor preservation conditions, which has caused some damage. As a result, parts of the text are missing or unrecognizable, and the original text of the ancient books has been digitized, its content has been mutilated and cannot be used for further study and research. Currently, the most common method of restoring old books relies on the manual restoration of experts, who fill in the missing text based on the document paper's effective area and their access to a vast library of historical materials. This allows the standard text of the text's individual characteristics to finish the text's structural reasoning [1]. As is to be expected, the intricacy and duration of these repair techniques make them challenging for experts to perform. Moreover, the scarcity of individuals specializing in the restoration of ancient books adds to the challenge of the restoration process.

This paper uses a deep learning model to automatically predict the missing text. The model takes the text sequence of the missing text as input and is used to predict the missing text after training, which can be used to help the professionals in restoration and to improve the efficiency and accuracy of the restoration. This approach hopes to decrease the difficulty of restoring ancient books, increase the efficiency of restoration, and ensure the integrity of the content of the digitized ancient books.

Particularly in light of the recent rapid development of pre-trained language models, which has further advanced the field of natural language processing and shown great promise for both understanding and producing natural language, deep learning language models have demonstrated impressive performance in a wide range of applications. While more powerful large language models are unsuitable for practical implementation due to their excessive demand for computational resources, there are fewer language models specifically trained for ancient Chinese, and their performance on the ancient Chinese dataset is not satisfactory. Thus, in this paper, the model architecture is redesigned to improve its performance in the task of predicting missing words in ancient texts, based on the pre-training model SikuRoBERTa [2], which is specifically designed for ancient Chinese language. According to the experimental findings, the single-word prediction accuracy on the dataset used in this study is 76.77 %, while the Top-5 accuracy is 90.66 %. These findings open up new avenues for the restoration of ancient books and offer compelling evidence in favor of more research into the digitization, processing, and restoration of ancient Chinese texts.

2 RELATED WORK

In recent years, several deep learning models have been proposed for text-filling tasks as a result of advancements in Natural Language Processing (NLP) technology. In order to solve the general task of text filling—filling any position and any number of missing texts—Zhu et al. [3] proposed the Mask-Self-attn model, which is based on the attention mechanism. The model produced the expected results on English texts. In 2019, Assasel et al. proposed the Pythia [4] model, which is the first deep learning neural network to be used for ancient text restoration. Pythia's codec is an LSTM network, and it contains an attention mechanism. The model performed well on the ancient Greek inscription dataset. The Blank Language Model (BLM), which Shen et al. [5] proposed, creates sequences by dynamically filling in the blanks. BLM is perfect for a range of text editing tasks because it uses the blanks to control the portion of the sequence to expand; To address the issue of text semantic fluency after filling, Tian et al. [6] proposed a predictive network for semantic fusion loss; Assasel et al. [7] proposed the Ithaca model in 2022 in response to the refinement of the Pythia model that had been previously proposed. Using the encoder structure of the Transformer, this model is the first to address the three primary tasks in a goldsmith's workflow holistically. It is also capable of accurately predicting the time, location, and missing text of ancient Greek inscription texts.

It is discovered that the majority of earlier work only concentrates on the filled text's quality in an attempt to produce a more semantically fluid text; it does not concentrate on enhancing the predicted text's accuracy. Additionally, the remaining ancient text restoration models do not target the ancient Chinese language specifically and do not achieve a satisfactory accuracy rate. Since most language models are unable to accurately capture the semantic features of ancient Chinese, the prediction of missing text does not produce the desired results.

3 ANCIENT CHINESE TEXT RESTORATION MODEL DESIGIN

The majority of today's ancient texts are written in traditional Chinese, and the vocabulary is rich and diverse. Many ancient words are no longer in use, and there is also a significant amount of proprietary vocabulary that does not exist in modern Chinese. Consequently, this paper builds a text restoration model for the ancient Chinese language in order to address these issues. The model performs exceptionally well in the text filling task and increases the prediction accuracy by guaranteeing the semantic fluency of the text. To aid ancient book restorers in their work, this paper's model also provides the top five predictions for the likelihood of a position being filled. This can significantly streamline the restoration process, increase productivity, and open up new avenues for future ancient book restoration endeavors.

The pre-trained SiKuRoBERTa model [2] served as the foundation for the design of the ancient Chinese text restoration model presented in this paper. The SiKuRoBERTa model's basic structure is still shared by the BERT model [8]. The model is a high-performance BERT that was obtained through a series of careful optimizations, and it combines a large amount of high-quality ancient text corpus to be trained on this structure. This allows the model to perform even better on many ancient text processing tasks because it has a more powerful feature extraction capability for ancient texts.

Figure 1 depicts the general architecture of the ancient Chinese text repair model, which can be loosely categorized into three sections: the input embedding layer, the encoding layer, and the prediction network. The input embedding layer converts each word of the input text into vector form and provides rich input information for the model. The encoding layer uses the feed-forward neural network structure and the multi-head self-attention mechanism to capture the semantic features and word relationships in the vectorized text sequences, learn the contextual context, and create the representation of each word in the context. The filler text is predicted by the prediction network, on the other hand, using the features that come before and after the missing position. It does this by interpreting the data that the coding layer has learned.

3.1 Input Embedding layer

In the conventional BERT architecture [8], the input embedding is comprised of three primary components: token embedding, positional embedding, and segment embedding. The token embedding serves as a representation of the smallest unit of text, specifically mapping Chinese characters into a vector space. This approach ensures that each Chinese character is assigned a distinct representation, thereby enhancing the model's ability to effectively capture both semantic and syntactic information within the textual data; Positional embedding can be conceptualized as a vector that encapsulates the positional information associated with each token embedding. The BERT model addresses the long-term dependency issues that are characteristic of recurrent neural networks and is capable of processing data in a parallel manner. However, this parallel processing also results in a lack of sequential relationships among the data, which significantly affects text processing. To mitigate this limitation, the BERT model incorporates positional embeddings, enabling the model to comprehend the semantic meaning of sentences and their grammatical structures by considering the order of lexical elements. This approach enhances the model's ability to capture the relative positional relationships among lexical elements within a sentence. The specific methodology for calculating positional encoding is detailed as follows:

$$PE(pos, 2i) = sin(\frac{pos}{10000^{\frac{2i}{d_{model}}}}) \tag{1}$$

$$PE(pos, 2i+1) = cos(\frac{pos}{10000^{\frac{2i+1}{d_{model}}}}) \tag{2}$$

where P E denotes the encoded vector of this positional information, pos denotes the position of the word in the sentence, i is the position

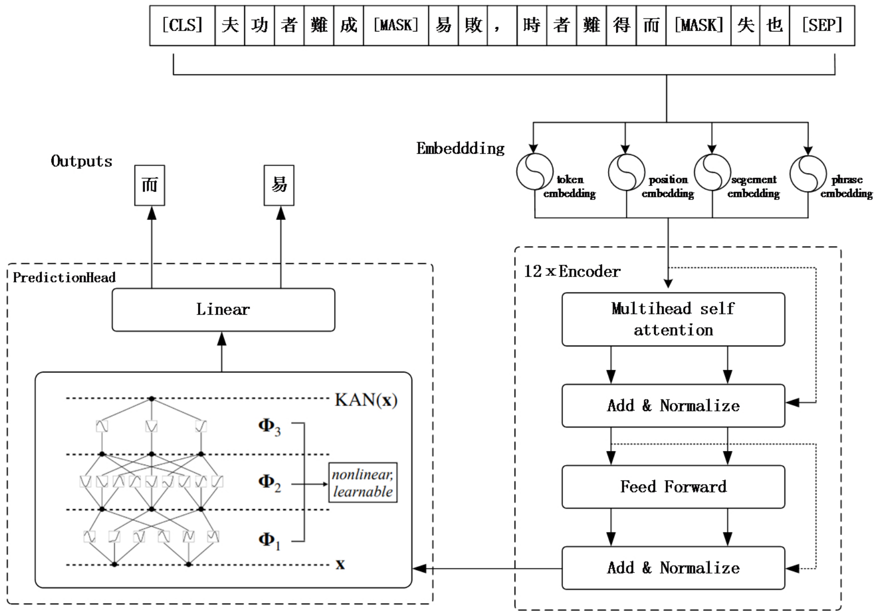

Figure 1: Overall model structure.

of the word vector, and d_{model} denotes the dimension of the word vector.

Segment embedding is employed by BERT to encode distinct sentences in specific natural language processing (NLP) tasks, facilitating the model's ability to discern the relationships among various sentences. By utilizing sentence embedding, BERT acquires both the semantic information contained within individual sentences and the semantic relationships that exist between different sentences, thereby enhancing its comprehension of textual input.

Nonetheless, there are certain shortcomings with the current BERT embedding input layer. By utilizing the BERT model, we can acquire an embedded representation of the text at the character level. However, Chinese data differs from other languages in that it is composed entirely of Chinese characters without spaces, and any combination of Chinese characters has the potential to form new words or phrases with distinct meanings. For instance, in historical writings, the phrase "通判" (general judgment) denotes a title or official position, such as that of a judge, which was one during the Ming and Qing dynasties. "通" or "判," as separate characters, have entirely distinct meanings. In the context of the Chinese language, phrase-level semantics is essential for understanding the overall semantics of a text, especially in the case of ancient Chinese, which exhibits a pronounced reliance on phrase-level meaning. Nevertheless, the conventional BERT model is limited in its capacity to capture phrase-level semantics, leading to a significant deficiency in its ability to accurately predict missing text. Consequently, motivated by the study of ZHAO et al. [9], the model in this paper keeps the three original BERT embedding layers and adds an average pooling layer to the input embedding to make token embeddings undergo average pooling. This allows the features of the token embedding data to be aggregated and compressed, and the most significant features are extracted to capture the semantic information at the phrase level, i.e. E. embeddings of phrases. Lastly, as illustrated in Figure 2, the original three input embeddings are linearly added to the phrase embeddings, enabling the model to derive semantics at both the character and phrase levels.

3.2 Encoder layer

The encoder layer is the core of the whole model, which can perform multi-level feature extraction, representation learning and context-awareness on the vectorized text sequence. In order to capture the semantic features embedded in the context, the model in this paper adopts the Multi-head self-attention mechanism in the Transformer structure, where the attention mechanism of each head is computed in parallel, and the outputs of each attention mechanism are spliced in the end, which enables the model to efficiently learn the textual data when processing the This allows the model to effectively learn sequence dependencies, global correlations and semantic information when processing text data. The multi-head attention mechanism projects "queries", "keys" and "values" to different representation subspaces Q, K, V through parameter matrices respectively, and calculates the corresponding attention. The attention calculation formula is:

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}}) \tag{3}$$

The ancient Chinese text restoration model presented in this paper employs a configuration of 12 encoder layers. Each encoder layer incorporates elements such as a multi-head attention mechanism and a feed-forward neural network. This deep network architecture enhances feature representation and increases the model's expressive capacity, thereby enabling effective handling of ancient texts characterized by intricate semantics.

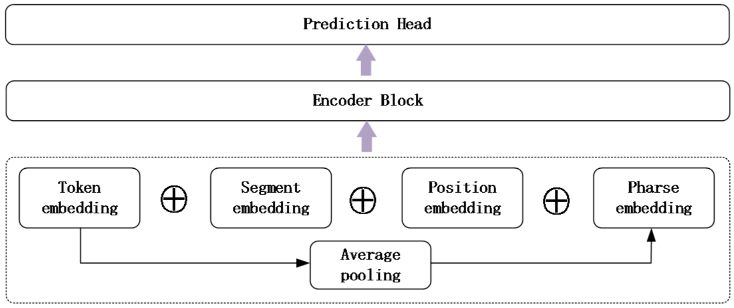

Figure 2: Model input embedding layer.

3.3 Prediction Network

The function of the prediction network is to convert the output feature vectors, which have undergone processing by the encoder, in order to estimate the conditional probability distribution of the masked words. The prediction layer within the BERT architecture is composed of two fully connected layers accompanied by the GELU activation function [10]. The output feature dimension of the final fully connected layer is configured to correspond to the vocabulary size, thereby facilitating the mapping of the hidden state to the score associated with each word in the vocabulary. The word with the highest score is identified as having the greatest probability of occupying the specified position. It is noteworthy that the original prediction network of BERT adheres to the Multi-Layer Perceptron (MLP) framework, which presents several limitations in practical applications. Specifically, the fixed nonlinear activation function inherent in the perceptron structure lacks the capacity for adaptive adjustment across varying tasks. Furthermore, as the network depth increases, it encounters challenges related to vanishing and exploding gradients. Additionally, the extensive number of linear combinations present within the MLP leads to a substantial increase in the number of parameters.

To enhance the predictive accuracy and address the limitations associated with traditional Multi-Layer Perceptrons (MLPs), we have restructured the prediction network utilized in the ancient Chinese text restoration model. This redesign incorporates the KAN (Kolmogorov-Arnold Networks) framework, drawing upon the concepts presented in the work of Liu et al. [11]. Conventional MLPs typically apply a nonlinear activation function subsequent to the linear combination of inputs, thereby facilitating the adaptation to and learning of intricate nonlinear relationships and data distributions. In contrast, KAN networks implement a direct nonlinear activation of the inputs prior to their combination. These networks are grounded in the Kolmogorov-Arnold representation theorem, which posits that a high-dimensional function can be effectively reduced to a one-dimensional function that captures polynomial orders of magnitude, as expressed in the following formulation:

$$f(X) = f(x_1, ..., x_2) = \sum_{q=1}^{2n+1} \Phi_q(\sum_{p=1}^{n} f_{p,q}(x_p)) \tag{4}$$

Where$\phi_{p,q}$: $[0, 1] \rightarrow \mathbb{R}$, Φ_q:$\mathbb{R} \rightarrow \mathbb{R}$. The structure is shown in Figure 3.

In the literature, the authors propose substituting the components of the weight parameter matrix in the multilayer perceptron (MLP) with B-spline functions, which serve as learnable activation functions characterized by variable sub-parameters. Concurrently, to facilitate the deep stacking of the network, the MLP is utilized as the "external skeleton," thereby integrating both nonlinear and linear operations within the network. Additionally, the network is augmented with spline functions that function as activation mechanisms.

Here, Φ_l is the B-spline matrix corresponding to layer l and x is the input matrix. It is worth noting that ϕ(x) in "(5)" is improved based on the B-spline function so that the activation function is ϕ(x) is the sum of the B-spline function and the basis function b(x), which is similar to the residual connection:

$$\phi(x) = w_b b(x) + w_s spline(x) \tag{5}$$

In most cases, spline(x) is parameterized as a linear combination of B-splines, and c_i is trainable in "(5)" defined as:

$$spline(x) = \sum_i c_i B_i(x) \tag{6}$$

The basis function is defined as:

$$b(x) = silu(x) = x \big/ \left(1 + e^{-x}\right) \tag{7}$$

The KAN network, due to its limited reliance on numerous linear combinations, possesses a reduced number of parameters in comparison to the Multi-Layer Perceptron (MLP) when both architectures are configured with an equivalent number of layers. Furthermore, the learnable nature of its activation function enhances its capacity to adaptively fit the final output curves for various tasks, thereby contributing to improved accuracy.

4 Parameter setting and experimental data

4.1 Problem Description

The model uses the unique symbol "[MASK]" to indicate the masked words in order to simulate the situation of missing words in damaged ancient books. We anticipate that the model will be able to correctly predict the words in the masked position and produce complete, semantically coherent, and meaningful text. Suppose the complete text to be input is s =(s_1,s_2, . . . ,s_{n-1},s_n) , n denotes the length of the text sequence. The $s_{template}$ is obtained by randomly masking the complete text s. Assuming that the positions

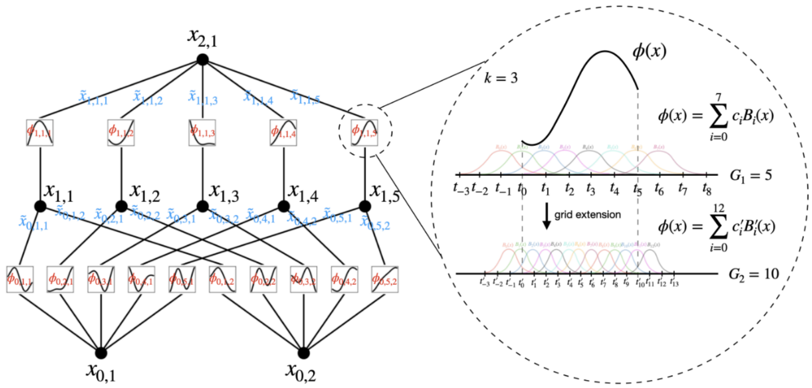

Figure 3: KAN network.

of the sequence u and the sequence v are masked in thes_template, and we use the simplified "[M]" to denote a masked identifier, $s_{template}$=(s_1,s_2, . . . ,s_{u-1},[M], . . . ,s_{v-1},[M], . . . ,s_n), and the masked are the labels.

4.2 Dataset Preparation

The ancient texts used in this paper are all in traditional Chinese and come from a total of 700 ancient texts in the fields of geography, science and technology, history, medicine, and other subjects. We obtained the TXT texts of these texts from publicly accessible online resources. To investigate various scenarios of text fragmentation in ancient manuscripts and to assess the efficacy of the proposed model for text reconstruction, this study developed two distinct datasets for experimental purposes.

The first dataset comprises a collection of ancient Chinese short texts, from which 300 texts were randomly selected and segmented into individual short texts, each containing a minimum of 30 words and a maximum of 60 words. The resulting dataset consists of 424,144 training texts and 47,350 test texts, with each text subjected to a random masking of one word.

The second dataset comprises an extensive collection of ancient Chinese texts, specifically consisting of the remaining 400 texts. These texts will be divided into segments, with each segment containing between 256 and 512 words. A portion of this text will be subjected to masking at varying proportions to evaluate the model's efficacy in reconstructing the content when a greater number of words are omitted.

4.3 Model parameter setting

The model presented in this paper is developed utilizing the deep learning framework PyTorch. Its primary objective is to forecast the probability distribution of a masked word by leveraging the surrounding context. This prediction can be effectively conceptualized as a binary classification task, where the outcome is either the masked word or an alternative. Consequently, the model employs a binary cross-entropy loss function.

The optimization method of the loss function is chosen Adam optimizer [12], and the exponential decay rate of the momentum term β_1, β_2 is 0.9, 0.99, ϵ =10-9, respectively. In this study, we opted not to utilize a fixed learning rate due to the potential drawbacks associated with both excessively small and excessively large values. A learning rate that is too small can lead to slower convergence of the model, while a learning rate that is too large may cause the model to become trapped in a local optimum, thereby hindering convergence. To address this issue, we implemented the learning rate optimization strategy provided by the cosine annealing learning rate (cosine annealingLR) algorithm. The cosine function exhibits a unique behavior where, as the variable x increases, the cosine value initially decreases gradually, followed by a more rapid decline, and subsequently slows down again.

4.4 Evaluation metrics

The evaluation metrics utilized in this study encompass "Top-k," which refers to the proportion of accurate predictions among the initial k outputs generated by the model. It is important to note that while the model presented in this paper is capable of rapidly generating predictions for missing texts, this does not imply that it can supplant the roles of historians and specialists in antiquities restoration. Rather, the objective of our model is to enhance the capabilities of antiquities restoration professionals and significantly increase the efficiency of their work. Consequently, this paper will assess not only the accuracy of the model's primary prediction but will also include an evaluation of Top-5, which measures the percentage of correct predictions found within the first five outputs relative to the test corpus.

Furthermore, this study employs the metric of perplexity to assess both the efficacy of the language model and the quality of the text produced.

5 Experimental evaluation and analysis

The model presented in this study is an enhancement of the preexisting SiKuRoBERTa model. To assess the efficacy of the proposed improvements, a decoupling methodology is employed during the experimental evaluation phase. This approach allows for the independent assessment of the roles and contributions of both the average pooling layer and the KAN network within the context of

[CLS]韓魏公與範文正公議西事不合，[MASK]正徑拂衣起去，魏公自後把住[SEP]
文 | 0.9999891519546509
武 | 6.4712548919487745e-06
公 | 2.8920107979502063e-06
解 | 8.424172506238392e-07
敬 | 7.737012879260874e-07
[CLS]范堯夫罷相，與伊川相見，責堯[MASK]曰：曩者，某事相公合言，何[SEP]
夫 | 0.9953147172927856
川 | 0.00406039971858263
臣 | 0.0005328747211024165
兩 | 5.9084934036945924e-05
連 | 3.298255251138471e-05
[CLS]新法之變，議者紛然。伯淳見介甫，介甫聞伯淳至，盛怒以待之。伯淳見介甫，介甫聞伯淳至，盛怒以待之。伯淳[MASK]見，和氣藹然見眉宇間，即笑謂介甫曰：今日諸公所爭，皆非私，實天下事。求相公少霽威色，且容大家商量。[SEP]
既 | 0.9994633793830872
初 | 0.0001662795402808115
若 | 0.00012732885079458356
叔 | 0.00012657578918151557
太 | 0.00011639043077593669

Figure 4: Test corpus excerpt.

Table 1: Short textbook filling experiment test results.

Models	Top-1acc(%)	Top-5acc(%)	PPL
Bert-wwm	57.2	70.6	9.97
S_RoBERTa	72.9	88.2	3.69
S_RoBERTa_AP	76.2	90.6	**3.16**
Ours	**76.8**	**90.7**	3.34
BART	32.6	53.9	57.42
Ithaca	68.1	85.6	5.02

the text-filling task. The experimental design incorporates ablation studies, which facilitate a comprehensive understanding of how various components of the model influence its overall performance. Also, the model proposed in this paper will be compared with the following baseline model:

- **BERT-wwm**: An improved version of BERT for the pre-trained model, which adopts the Chinese Whole Word Mask [13] (WHOLE WORD MASK), which is more suitable for Chinese text learning than BERT.
- **BART** [14]: A model based on the Transformer encoder and decoder, which utilizes the encoder to extract semantic features and the decoder to generate the filled text.
- **Ithaca**: This is a model proposed by the Google team for the prediction of missing text in the text of ancient Greek inscriptions, which introduces the Bigbird attention mechanism and is based on the Transformer encoder.

5.1 Experimental evaluation and analysis of text filling

We first evaluate the ability of each model to predict missing text separately by testing it on the ancient Chinese short text dataset. The Top-5 predictions of the models in this paper are shown in Figure 4. The models rank the five predictions given in order of their scores.

The results of the post-experimental evaluation of each model are shown in Table 1.

The experimental findings indicate that the model presented in this study outperforms other models in terms of prediction accuracy and overall performance. This superiority can be attributed to the model's consideration of semantic information at the phrase level, in contrast to the Ithaca model and the SiKuRoBERTa model, which focus solely on character and word-level semantics. Furthermore, the model in this study is capable of capturing richer semantic relationships among multiple adjacent characters within the text, thereby enhancing its comprehension of deeper semantic meanings. Additionally, the KAN network utilized in the prediction layer aligns

Table 2: Long Text Fill Experimental Test Results.

Mask-rate	Metrics	S_RoBERTa	S_RoBERTa_AP	BERT-wwm	Ours
10%	Top-5 acc	74.31	75.46	64.18	**75.72**
	PPL	10.43	9.33	40.16	**9.08**
20%	Top-5 acc	69.61	70.83	60.70	**71.62**
	PPL	14.28	12.91	49.01	**12.06**
30%	Top-5 acc	64.75	65.59	56.41	**66.91**
	PPL	20.05	18.93	63.12	**16.63**
40%	Top-5 acc	59.05	59.63	52.01	**61.50**
	PPL	30.36	29.63	83.18	**24.07**

more effectively with the probability distribution of predictions, leveraging the information derived from the hidden layer of the preceding network, which contributes to improved accuracy during the prediction phase. And the BART model performs even worse, even though it consists of an encoder and a decoder together, but BART is also not applicable to the task of this paper due to different training purposes.

Subsequently, we will evaluate each model's performance using the ancient Chinese long text dataset and see how well it fills in under various mask ratios. Table 2 compares the experimental outcomes for each model.

Table 2 presents the evaluation results of the various models. It is clear from these that as masking rates rise, it becomes harder to predict and fill in the missing text. Additionally, prediction accuracy and model performance decline because the model is able to extract fewer contextual semantic features from the text due to insufficient contextual information. While the lack of contextual information on this model is not as severe as it is on other models, the model in this paper can still function well when there is a greater amount of missing contextual semantic information. This is because our model can also obtain phrase-level semantics. The experiments above also demonstrate that our model can be applied to the task of restoring ancient texts, better supporting experts in the field.

6 conclusion

In this paper, we propose a model for the restoration of ancient Chinese texts based on the pre-training model RoBERTa. The model makes better use of the hidden information from the upper layers to reduce loss and enhance prediction ability; the prediction layer based on KAN network ensures that the model can capture the semantics of multiple neighboring characters during the training process, enhancing the ability to understand the contextual semantics. The ablation property of our experimental evaluation validates the efficacy of the improvement method presented in this paper for the pre-trained model SiKuRoBERTa. In the meantime, comparison with the other baseline models demonstrates that the model presented in this paper is more capable of completing the task of restoring ancient book texts. It can also be implemented into the ancient book restoration workflow, which can enhance the accuracy and efficiency of the restoration process.

References

[1] Chen Shan-Xiong, Zhu Shi-Yu, Xiong Hai-Ling, *et al.* **2022**. A method of inpainting ancient Yi characters based on dual discriminator generative adversarial networks. Acta Automatica Sinica, 48(3): 853−864.

[2] Wang Dongbo, Liu Chang, Zhu Zihe, *et al.* **2022**.Construction and Application of Pre-trained Models of Siku Quanshu in Orientation to Digital Humanities[J]. Library Tribune, 42(06): 31-43.

[3] Zhu W, Hu Z, Xing E. **2019**.Text infilling[J]. arXiv: Computation and Language.

[4] Assael Y, Sommerschield T, Prag J. **2019.**Restoring ancient text using deep learning: a case study on Greek epigraphy[J]. EMNLP/IJCNLP: 6369-6376.

[5] Shen T, Quach V, Barzilay R, *et al.* **2020**. Blank language models[J]. EMNLP.

[6] Zheng S, Tian W, Cai X. **2020**. Text Infilling Method based on Key Semantic Information Selection Mechanism. 2020 5th International Conference on Information Science, Computer Technology and Transportation (ISCTT). IEEE: 219-223.

[7] Assael Y, Sommerschield T, Shillingford B, *et al.* **2022**. Restoring and attributing ancient texts using deep neural networks[J]. Nature, 603(7900): 280-283.

[8] Devlin J, Chang M W, Lee K, *et al.* **2019**. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding[J]. NAACL-HLT: 4171-4186.

[9] Zhao S, Zhang T, Hu M, *et al.* **2022**. AP-BERT: enhanced pre-trained model through average pooling[J]. Applied Intelligence, 52(14): 15929-15937.

[10] Hendrycks D, Gimpel K. **2016**. Gaussian error linear units (gelus)[J]. arXiv preprint arXiv:1606.08415.

[11] Liu Z, Wang Y, Vaidya S, *et al.* **2024**. Kan: Kolmogorov-arnold networks[J]. arXiv preprint arXiv:2404. 19756.

[12] Zhang Z. **2018**. Improved adam optimizer for deep neural networks. 26th international symposium on quality of service (IWQoS). IEEE/ACM: 1-2.

[13] Cui Y, Che W, Liu T, *et al.* **2021**.Pre-training with whole word masking for chinese bert[J]. IEEE/ACM Transactions on Audio, Speech, and Language Processing, 29: 3504-3514.

[14] Lewis M, Liu Y, Goyal N, *et al.* **2020**. BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension[C]. 58th Annual Meeting of the Association for Computational Linguistics. Association for Computational Linguistics: 7871.